DNA as a universal chemical substrate for computing and data storage

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Review article

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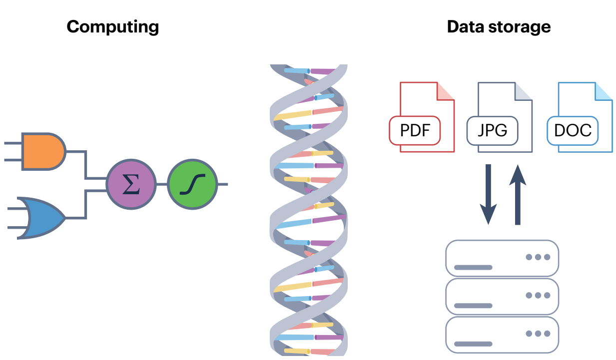
Abstract

DNA computing and DNA data storage are emerging fields that are unlocking new possibilities in information technology and diagnostics. These approaches use DNA molecules as a computing substrate or a storage medium, offering nanoscale compactness and operation in unconventional media (including aqueous solutions, water-in-oil microemulsions and self-assembled membranized compartments) for applications beyond traditional silicon-based computing systems. To build a functional DNA computer that can process and store molecular information necessitates the continued development of strategies for computing and data storage, as well as bridging the gap between these fields. In this Review, we explore how DNA can be leveraged in the context of DNA computing with a focus on neural networks and compartmentalized DNA circuits. We also discuss emerging approaches to the storage of data in DNA and associated topics such as the writing, reading, retrieval and post-synthesis editing of DNA-encoded data. Finally, we provide insights into how DNA computing can be integrated with DNA data storage and explore the use of DNA for near-memory computing for future information technology and health analysis applications.



Sections

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Introduction

Living systems, composed of physically confined and chemically controlled autonomous agents, demonstrate computational and learning capabilities that have thus far proved difficult to emulate using classical silicon-based devices. Moreover, interfacing silicon-based devices directly with processes in live cells remains challenging. Molecular computing, in which chemical species are used for computing, is an attractive alternative for operation in biological and chemical contexts that are incompatible with traditional electronics used in molecular diagnostics, data storage and information security[1,2]. In biology, information is encoded in DNA, which is a biopolymer that comprises sequences of four nucleotides – adenine (A), thymine (T), cytosine (C) and guanine (G) – connected by a deoxyribose and phosphate backbone. Nucleotides typically interact through Watson–Crick base pairing between A–T and C–G pairs[3] (Fig. 1a). Duplex formation (hybridization) between a pair of sequences in solution can be controlled by the temperature, pH and ionic strength of the solution[4] and accurately predicted using nearest-neighbour thermodynamic models[5]. This programmable and predictable nature of base pairing is essential for the construction of large-scale chemical reaction networks made of DNA[6].

DNA emerged as an alternative substrate for computing in 1994, when Adleman demonstrated that the travelling salesman problem could be solved using massively parallel operations of DNA reaction networks[7]. Two years later, Davis demonstrated that a visual icon could be converted into DNA sequences and stored in bacteria[8], indicating the possibility of using DNA as a medium for data storage. Compared with other biomolecules, such as proteins and sugars, or synthetic polymers, DNA is preferred as a computing substrate owing to its predictable and specific base pairing and to the maturity of the chemical and biological industry to support future developments. Although there are several approaches to constructing DNA-based computers, since being proposed by Yurke et al.[9] in 2000, DNA strand displacement (DSD) has become one of the most common methods for DNA computing owing to its simplicity[10,11]. The implementation of enzymatic reactions has expanded the toolbox for engineering DNA computational systems[12–16]. In addition, DNA self-assembled nanostructures such as tiles[17] and origamis[18] have been used to design algorithms for computing tasks[19–21], and nanomaterials, ranging from inorganic substances to biomolecules[22], have also been harnessed to facilitate DNA computation. In comparison with DNA computing, the development of DNA data storage has been slower. Since the 1960s, the natural information-carrying function of DNA led researchers to speculate on how it could be used as a data-storage medium[23,24], but it was not until 2012 that practical and scalable DNA data storage become a reality[25,26]. Information has been encoded in the structure of DNA[27,28], but a more widely used method is to encode information in the nucleotide sequence[29,30], much like nature does.

Although DNA computing and DNA data storage are at different stages of development, the field of DNA nanotechnology has facilitated the integration of these fields with the goal of developing a DNA-based computing architecture that can sense, process and store a large number of molecular inputs. This Review provides an overview of DNA-based computing and data storage for information technology and in vitro diagnostics. First, we summarize advances in DNA computing, including developments in logic circuits and neural networks based on enzyme-free DNA circuits, enzymatic DNA circuits, DNA nanostructures combined with nanomaterials and more complex compartment-based systems (Fig. 1b). Next, we discuss the current state of DNA-based data storage by outlining how digital data are encoded into DNA molecules, approaches for reading DNA archives, the targeted retrieval of data from larger pools and methods for the post-synthesis editing of DNA-encoded data (Fig. 1c). Finally, we discuss the possible integration of DNA computing and data storage as well as challenges remaining in the field.

Advances in DNA computing

Silicon-based computing performs diverse tasks using electronic logic gates, which receive and process an electronic input to produce an output signal. Such logic gates have been replicated using nucleic acids as signal-processing substrates. DNA-based neural networks have become important architectures in DNA computing, drawing on the successes of brain-inspired artificial neural networks to demonstrate pattern recognition and decision making in DNA-based systems. To achieve these diverse functions, various molecular toolboxes have been established, ranging from DNA-only systems and enzyme-assisted DNA reactions to DNA nanostructures and compartmentalized DNA circuits. Each of these approaches is discussed in more detail in the following sections.

Enzyme-free circuits

On a circuit level, the need for predictable and programmable behaviour has resulted in the development of toehold-mediated DSD as a common building block for constructing connected circuit layers, neural networks or compartmentalized circuits in aqueous media (Fig. 1b). Toehold-mediated DSD is the process by which a single-stranded DNA (ssDNA) molecule (referred to as the input) replaces an incumbent strand of a partially complementary pre-hybridized double-stranded complex (referred to as the gate)[31]. The reaction is initiated at a single-stranded toehold domain of the gate complex and progresses through branch migration, followed by a series of reversible single-nucleotide association and dissociation steps, resulting in the eventual release of the incumbent strand as an output. The reaction kinetics of DSD can be quantitatively modulated over approximately six orders of magnitude by changing the length and sequence of the toehold[4,32]. Combining these simple and effective primitives enables DSD to perform the same fundamental operations as electronic circuits, such as Boolean logic[10,33], arithmetic operations[34] and neural network functions[35].

Increasingly, there is a trend towards the development of more complex DNA computational mechanisms. For example, mechanisms inspired by electrical engineering, such as error correction[36] and switching circuits[37], have been adapted in DNA circuits. One error correction strategy involves making leak reactions (that is, unwanted displacement reactions) thermodynamically unfavourable by introducing an entropic penalty for these reactions using redundant sequences[36]. To emulate the switching circuits found in telecommunication, DNA switches based on DSD have been used as a standardized and basic functional element, enabling a DNA circuit to perform arbitrary Boolean computation with high modularity and speed[37]. Low-affinity partially complementary interactions are alternatives to typical DSD for DNA computing[38]. Although the kinetics are yet to be studied and predicted accurately, the higher reversibility of weaker pairing gives rise to faster strand exchange for faster computation and enables a specific input strand to bind to multiple partially complementary strands for high scalability. Automated design and machine learning have been used to investigate the performance of DNA computing with unpurified DNA strands[39] or in random-sequence pools with molecular noise[40]. These examples pave the way to the design of DNA circuits that are robust to molecular noise. As the complexity of DNA circuits has increased, high-level dynamic behaviours have

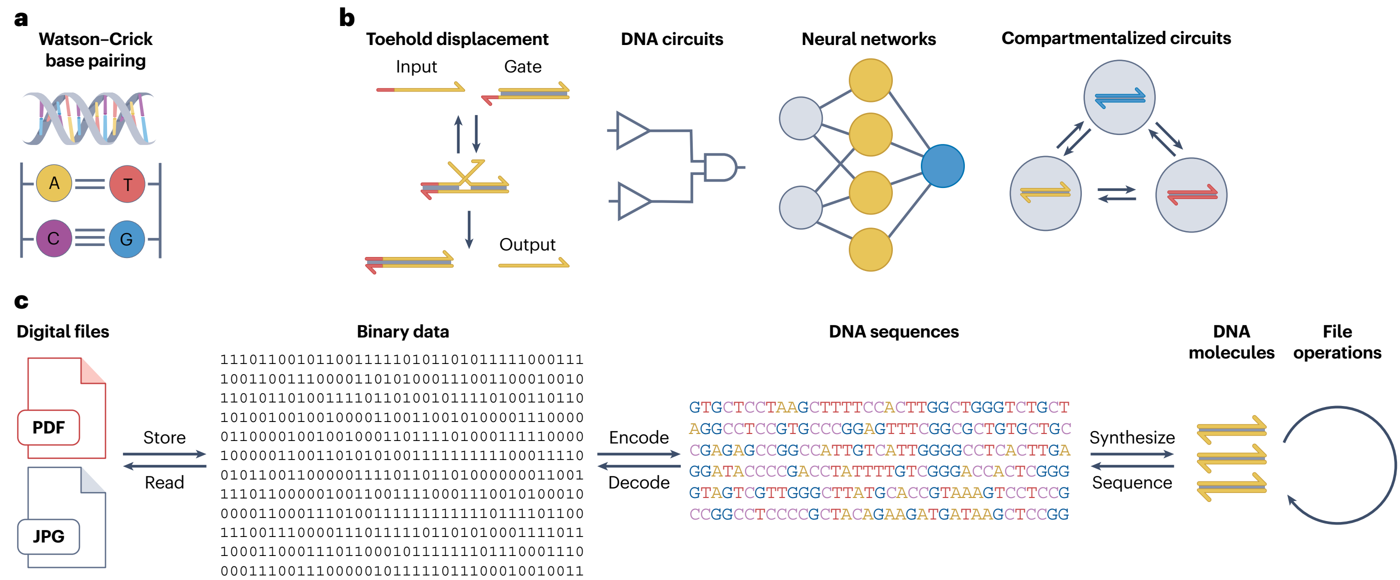

Fig. 1 | DNA as a substrate for computing and data storage. a, Base pairing enables DNA to serve as a substrate for molecular information processing. **b**, DNA strand displacement as a basic tool for logic gates, neural networks and compartmentalized circuits. In a typical DNA strand displacement, the single-stranded DNA input binds to the double-stranded DNA gate at the toehold domain (red) and initiates displacement. Complete displacement results in release of the output. The output can serve as an input to a downstream reaction, which allows for the connection of multiple displacement reactions and the construction of DNA circuits for DNA computing, such as logic gates and neural networks. In addition, compartmentalization offers a strategy to reach a higher level of complexity in DNA computing. **c**, DNA can be used to store digital data. Digital files such as text and pictures can be represented as sequences of bits and then converted into DNA sequences. The synthesized DNA sequences function as an operatable storage medium, which can be read by sequencing techniques and decoded back to digital data.

been engineered, including oscillations[41], probability switching[42], buffering[43] and temporal control[44], which demonstrate the possibility of building autonomous molecular systems with DNA as programmable matter. Complex circuits can be visualized and analysed using a software tool, such as VisualDSD[45] by compiling DSD reactions into a chemical reaction network that performs stochastic and deterministic simulations.

In addition to logic and arithmetic computation, DNA reaction networks have been programmed to function as molecular neural networks, inspired by the success of neural networks in the field of artificial intelligence[46,47]. The first experimentally demonstrated DNA-based neural network was used to recall ssDNA patterns through seesaw gate-based logic circuits[35]. This DNA-based perceptron is a DNA reaction network that calculates the weighted sum of the DNA inputs and returns a DNA output as defined by a threshold function, similar to perceptrons used in artificial neural networks inspired by the seminal work of Rosenblatt[48]. Using a Hopfield neural network strategy, the single-layer neural network consisting of four interconnected artificial neurons remembered four different patterns. Through the addition of specific DNA strands as 'clues', the neural network was able to recall the corresponding pattern. Extension of this molecular system through use of a more powerful winner-take-all neural network resulted in DNA neural networks capable of recognizing nine molecular patterns, with each pattern consisting of 20 distinct DNA molecules[49]. In a more recent development, convolutional neural network algorithms were implemented with synthetic DNA circuits based on a simple switching gate architecture[50] (Fig. 2a). Compared with perceptrons, the lower connectivity and complexity of the convolutional neural networks, which comprised a total of 368–512 distinct molecules, enabled the recognition of up to 32 molecular patterns, thus providing a pathway for classifying complex and noisy information with high computing power.

The pattern-recognition properties of DNA-based neural networks have been used in disease diagnostics and profiling. Owing to the low sensitivity of DNA classifiers (nanomolar concentrations) and the low concentration of biomarkers (femtomolar concentration range), preamplification is typically used to convert low concentrations of biomarkers into DNA or RNA strands with higher concentrations feasible for DNA classification. In one example of the application of a DNA classifier, a molecular computing strategy was used to analyse the information contained in gene expression signatures[51]. After training a computational classifier on publicly available gene expression data, the weights and other characteristics of the in silico classifier were translated into DNA complexes that replicated these functions in vitro. Experimentally, the DNA probes of the classifier were then coupled with a DNA-based circuit that enabled the detection of nonlinear-amplified RNA targets to distinguish between bacterial and viral infections. In a related approach[52], a computing platform for cancer diagnosis was used to analyse microRNA profiles in clinical serum samples. Similarly, the classifier was first trained in silico and then converted into a set of DNA complexes. In contrast to the nonlinear amplification of biomarkers in the previous example, a close-to-linear amplification method was used to catalytically produce DNA products without perturbing the concentration and variability information in the original samples. Addition of the amplified products to the classifier enabled accurate cancer diagnosis through a computation series consisting of multiplication, summation, subtraction and finally reporting of the outcome (Fig. 2b).

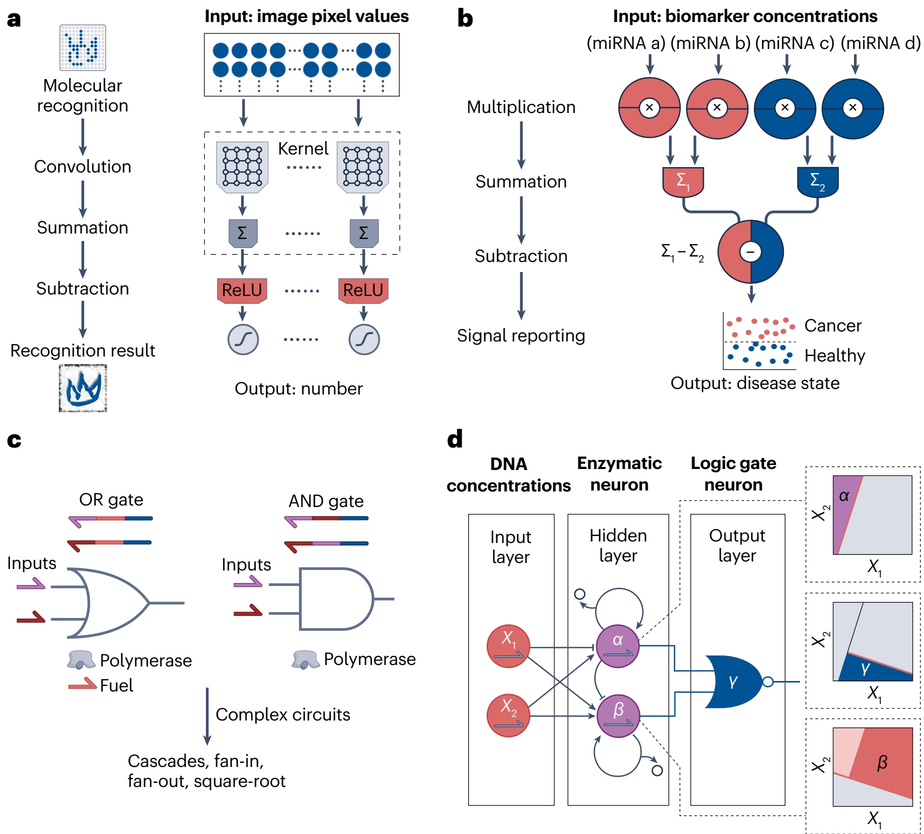

Fig. 2 | DNA logic gates and neural networks. a, Convolutional neural networks implemented with synthetic DNA circuits enable the recognition of handwritten symbols. The inputs of image pixel values are transformed by the convolution kernel, and then interact sequentially with downstream summation and subtraction gates to finally generate a fluorescence signal. The intensity of the signal is normalized to an output number as a score for pattern recognition. The rectified linear unit (ReLU) is a nonlinear activation function that converts negative values to 0. **b**, A DNA-based winner-take-all neural network enables the detection of microRNA (miRNA) for cancer diagnosis. The miRNA inputs are multiplied, summated and subtracted sequentially, resulting in the report of a value to distinguish the disease states. **c**, DNA logic gates that use strand-displacement polymerase. The hybridization of inputs and single-stranded gates results in the polymerase-mediated synthesis of new strands and concomitant strand-displacement reactions, which finally results in the release of output strands from the gate complex. Individual gates can be combined by cascading strategies (that is, sequences of consecutive chemical transformations) to construct large-scale logic circuits. Examples of such circuits include fan-in circuits (which process multiple inputs into a single output), fan-out circuits (which process a single input into multiple outputs) and circuits that use square-root functions to compute four-bit input numbers. **d**, An enzymatic neural network with two layers that can be used to classify complex mixtures of DNA by nonlinear decision making. The concentrations of two DNA inputs (X_1 and X_2) are detected by enzymatic reaction networks (α and β), the results of which are integrated by a third reaction network γ, resulting in classification of the input concentrations into three areas (shown in the dashed boxes). Part **a** adapted from ref. 50, Springer Nature Limited. Part **b** adapted from ref. 52, Springer Nature Limited. Part **c** adapted from ref. 16, Springer Nature Limited. Part **d** adapted from ref. 85, Springer Nature Limited.

These platforms could be further enhanced by developing automatic design and operation systems. Such an automatic platform for DNA classification has potential for use in large-scale data computing and medical diagnosis[53].

DNA neural networks could be further developed by drawing additional inspiration from silicon-based neural networks. The current generation of silicon-based neural networks relies on training data to learn and increase their accuracy over time. By contrast, the current generation of DNA neural networks are trained in silico but do not display learning behaviour over time. Thus, the next key step in the development of DNA neural networks is to implement adaptive DNA neural networks capable of learning. In most implementations of DNA-based neural networks, attempts have been made to translate the success of globally trainable networks to the molecular level[54–56]. Alternatively, physical learning[57,58] might provide an easier route towards self-learning circuits that might be more compatible with molecular computing. In physical learning, the network nodes learn locally in response to stimuli and do not require the full circuit to be simultaneously updated. Additionally, the learning and prediction steps do not have to be strictly separated, as is the case for globally updated networks. However, irrespective of the chosen learning approach, there are three main challenges that need to be addressed before learning of DNA neural networks can be achieved. First, the molecular species consumed during the learning process will need to be restored, which might require the use of enzymes or methods to resupply lost species through compartmentalization, microfluidics or auxiliary species. The second challenge will be to endow DNA-encoded data with a circuit-accessible domain that can be used for training. Finally, crosstalk between circuit components should ideally be reduced as much as possible; again, this might be achievable by localizing components that could potentially interact inside separate compartments and careful sequence design[41].

Enzyme-assisted DNA circuits

In bottom-up synthetic biology, DNA reaction networks are often constructed using simple enzymatic reactions with high specificity and superior catalytic ability compared with DNA-only systems[14,15]. Polymerase-mediated DSD has enabled highly efficient computation with simpler circuit structures and faster computing speed compared with computation based on enzyme-free DSD[16,59]. One of the main challenges in DNA circuits is leakage, which is assumed to be caused by imperfect sequence design and imperfectly hybridized multistranded complexes owing to synthesis errors. But as polymerase-mediated DSD allows for the use of simpler structures as gate complexes, it mitigates the problem of leakage[16] (Fig. 2c). Methods have been developed for the prescribed synthesis of ssDNA[60] and RNA circuits[61–63] that use nucleic acid biomarkers as inputs and execute computation through enzymatic reactions, which could enable the circuits to be interfaced with biological systems for potential use in diagnostics and biosensing. The stable performance of enzyme-assisted DNA circuits requires consistent enzymatic activity, which can be achieved by carefully controlling the batch quality, buffer conditions and temperature.

Besides polymerases, a particularly promising class of enzymes for DNA computing are Cas nucleases of the clustered regularly interspaced short palindromic repeats (CRISPR–Cas) system[64]. In this system, Cas nucleases are bound to a guide RNA that has complementary bases to, and thus recognizes, the DNA targets. Therefore, rational design of complementary guide RNA enables various applications, such as DSD-controlled Cas nuclease activity to regulate gene expression in live cells[65] and CRISPR-responsive hydrogels for in vitro applications, including tissue engineering, bioelectronics and diagnostics[66]. Specific Cas proteins exhibit indiscriminate nuclease activity after target binding, which results in the unselective cleavage of all single-stranded nucleic acids. By taking advantage of this indiscriminate catalytic ability, Cas proteins have been adapted to activate fluorescent single-stranded DNA reporters and integrated with DNA circuits for sensitive molecular diagnostics[67,68]. To increase the application scope of Cas proteins beyond catalytic signal amplification, Cas-based DNA circuits have been developed that use a partially catalytically inactive Cas protein to cleave specific strands[69]. Alternatively, DNA with catalytic properties can function as an enzymatic species. Typically, a DNAzyme is an ssDNA structure that recognizes target nucleic acids through base pairing and subsequently performs cleavage or ligation[70]. Taking advantage of their catalytic properties, DNAzymes have been used to construct logic gates and to control DNA circuits[71–73]. However, typical reaction rates of DNAzymes are much lower than those of protein-based enzymes.

Compared with circuits based on a single enzyme, DNA circuits based on multiple enzymes might have advantages owing to an increase in the range of dynamic and computational behaviours. One example is the genelet toolbox in which the control elements, genelets[14], are constructed from double-stranded DNA templates that contain an incomplete T7 RNA polymerase promoter region, from which transcription is severely inhibited. Activation of a transcriptional switch is achieved when an ssDNA activator hybridizes with the single-stranded region of the genelet, thereby completing the RNA polymerase promoter region and turning on transcription. The transcribed RNA in turn acts as an excitatory or inhibitory regulator for the transcription of other genelets by controlling the availability of the ssDNA activator of the target genelet. In addition to signal production by transcription, RNA signals are degraded by ribonuclease H. The combination of signal production and degradation establishes a signal turnover process as an out-of-equilibrium steady-state activity[74,75], through which active molecular systems with life-like features could emerge[76]. The polymerase, exonuclease and nickase (PEN) toolbox uses a similar strategy to construct enzymatic DNA reaction networks[15]. Short ssDNA molecules serve as inputs for ssDNA template strands, and extension of the inputs by a DNA polymerase produces fully complementary double-stranded DNA complexes. Subsequently, the extended DNA strand is cut by a nickase and both fragments dissociate from the template strand. The template strand can also be deactivated by an ssDNA inhibition strand that competes with the activator for template association but has a mismatch at the 3′-end, which prevents extension of the partial duplexes. Finally, activator, inhibitor and product strands can be degraded over time by an exonuclease. The use of polymerase, exonuclease and nickase permits autocatalytic, linear and inhibitory reactions to be interconnected, resulting in networks with sophisticated spatiotemporal properties[77,78]. Additionally, more complex genetic circuits termed in vitro transcription and translation (TXTL) reactions[79] have been used as a molecular toolbox to construct synthetic circuits[80,81]. Collectively, these enzymatic DNA networks have demonstrated great potential for the realization of diverse dynamic and out-of-equilibrium behaviours, which have enabled them to perform complex computational tasks. One of the challenges for these enzymatic DNA networks will be to demonstrate scalability beyond relatively few network layers, as scaling of the number of layers leads to an increase in nonlinearities such as retroactivity and resource competition[82–84].

Neural networks can be engineered by enzymatic DNA reactions to achieve higher sensitivity[56,85] (Fig. 2d) than those constructed using only DNA species. One example is a PEN toolbox-based neural network that displays nonlinear decision-making capabilities[85]. The neurons are enzymatic DNA reaction networks that mimic the nonlinear activation function of perceptrons and can detect RNA at picomolar concentrations and discriminate between inputs for which concentrations differ by only 10–20%. Three neurons connected into a two-layer neural architecture recognized the concentration signatures of more complex RNA mixtures. This system could be extended by increasing the number of layers to enable the processing of larger-scale data from real-world applications. Finally, more complex TXTL reactions have been used to construct a basic perceptron[86]. The weighted-sum operation uses toehold switch riboregulators[81] and the green fluorescent protein output is translated only when the upstream product sigma factor exceeds the threshold set by an anti-factor. Although tuning of the system is delicate because of the inherent variability in the transcription and translation between different input and weight pairs, TXTL-based computation might provide additional autonomous learning abilities by enabling the continuous training of neural networks and allow for the construction of larger complex synthetic genetic networks that are compatible with various inputs and outputs, such as proteins and small molecules.

DNA nanostructures

Structural DNA nanotechnology started with the seminal work of Ned Seeman and the construction of immobile Holliday junctions[17]. This work was subsequently extended to the construction of DNA tiles – DNA nanostructures consisting of branched junctions – that serve as basic building blocks for the self-assembly of higher-order structures[22]. Over the

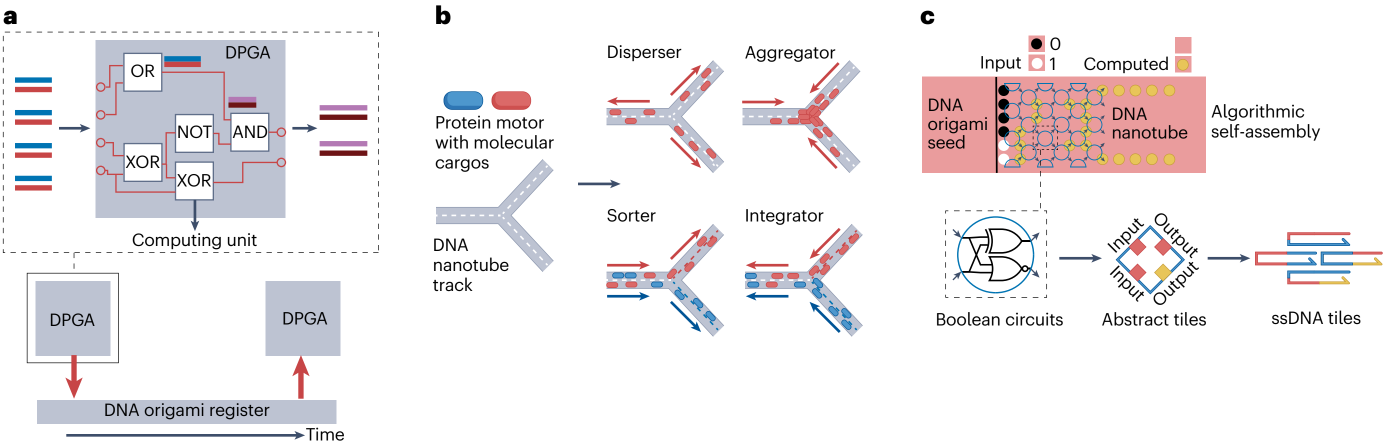

Fig. 3 | DNA nanostructures for DNA computing. a, DNA-based programmable gate arrays (DPGAs). DNA inputs (light red and blue strands) are processed by one DPGA containing several subcircuits (for example, OR, AND, NOT and XOR). The generated outputs (pink and dark red strands) are written to the DNA origami register by strand displacement and then transmitted to a downstream DPGA. Integration of multiple DPGAs enables general-purpose DNA computing. **b**, Protein-based motors are designed to move on DNA nanotube tracks for molecular transport tasks, enabling the construction of a disperser that moves cargos (blue and red dots) from the centre to the periphery of the DNA track, an aggregator that moves cargos from the periphery to the centre of the track, a sorter that splits the cargo stream into two routes and an integrator that converges two cargo streams into one route. The red and blue arrows indicate the direction of movement of cargos of that colour. **c**, Single-stranded tiles can be programmed to execute algorithms. The iterated Boolean circuits are abstracted to a tile-assembly model, which can then be translated to DNA sequences. Upon input addition, the tiles self-assemble on the DNA origami seed to perform computation. ssDNA, single-stranded DNA. Part **a** is adapted from ref. 93, Springer Nature Limited. Part **b** is adapted with permission from ref. 94, American Association for the Advancement of Science. Part **c** is adapted from ref. 21, Springer Nature Limited.

past four decades, these initial tiles have been improved to enable the construction of DNA structures with controlled geometry, topology and connectivity[22]. DNA self-assembly has been further advanced by the development of DNA origami, which is a robust and efficient technique for making rationally designed shapes[18]. DNA origami relies on the use of an ssDNA scaffold that is folded with hundreds of short staple strands into DNA nanostructures. Multiple DNA origamis have been designed before further assembly into micrometre-scale structures[87,88], which has taken structural DNA nanotechnology into the microscopic world. In this section, we discuss how DNA nanostructures have been used as scaffolds and units for DNA computing.

Compared with DNA molecules, a major advantage of DNA nanostructures is the possibility to predetermine the spatial organization of DNA circuits[20] to produce a functional and modular scaffold for solving specific tasks such as cargo sorting[89], maze solving[90], the construction of finite-state machines[91] and cryptography[92]. The surface addressability inherent to DNA nanostructures has been exploited to develop DNA origami registers that can temporarily store intermediate data and operate cascaded sub-circuits asynchronously, thereby increasing the scale and circuit depth of liquid-phase DNA digital computation[93] (Fig. 3a). In another example of DNA nanostructures being used as a scaffold, researchers used a hybrid approach to transport cargo on DNA nanotubes, leveraging the high specificity and efficiency of protein motors and the programmability of DNA nanostructures[94] (Fig. 3b). The protein motor was modified with DNA-binding domains, which enabled motion on the DNA tracks with average speeds of 200 nm s^{-1}, which is close to those of biological counterparts such as mitotic motors. By controlling the orientation of the recognition sequence on the track, unidirectional motion was used to perform tasks, namely, to disperse, aggregate, sort or integrate molecular cargos. The combination of motile elements, such as DNA tracks, directed protein motors and computational DNA circuits hints at a path towards smart molecular assemblies. Interaction of a DNA circuit with environmental cues and subsequent algorithmic updating of a molecular assembly line could, for example, allow for adaptive chemical synthesis in response to stimuli[95,96].

Alternatively to serving as scaffolds, DNA nanostructures themselves can serve as units for DNA computing. Reconfigurable DNA nanostructures can be programmed to interact with biomolecules and to make decisions based on Boolean logic in live cells both in vitro[97–99] and in vivo[100,101]. Further information on DNA nanotechnology for cellular applications can be found elsewhere[102–104]. As well as programming individual DNA nanostructures, DNA circuits can also organize and modulate multiple nanostructures to construct complex self-assembled networks for DNA computing. For example, toehold-free DSD[105] and DNA tile displacement[106] have been used for computation between interacting DNA nanostructures, enabling the nanostructures to be updated dynamically. To further optimize the diversity and robustness of DNA computing, DNA nanostructures have been used for various algorithmic, assembly-based logic operations[21] (Fig. 3c). Algorithmic information was encoded in a DNA origami seed that directed the growth of single-stranded tiles into nanotubes with precisely controlled patterns as the visible output. Careful sequence design is required to ensure the assembly of single-stranded tiles occurs from the origami seeds exclusively. However, this design principle also suggests that the scaling of the number of single-stranded tiles is constrained by the compatibility of the sequence design with the origami-directed seeding. In principle, such algorithmic assembly could be interfaced with systems based on RNA and proteins, which could enable the formation of dynamic structures[107,108]. Excitingly, the ability of DNA nanostructures to perform mechanical work[109] also hints at a future in which molecular machines with sub-micrometre dimensions are constructed from self-regulating and interacting DNA nanostructures. An attractive way to achieve such responsive nanostructures could be through the construction of mechanical neural networks[110,111] based on DNA. Mechanical neural networks are based on interlinked connectors that can change their properties analogous to the weights of a neuron in an artificial neural network. Using DNA origami, it is relatively straightforward to build complex structures from reusable components with programmable mechanical properties[112,113]. Updating the properties of individual components could be realized using DNA metamaterials[114] or toehold-mediated DSD[115].

Bead-based and compartmentalized DNA circuits

In addition to DNA nanostructures, other materials, such as colloids, lipids, glass substrates and synthetic compartments, might offer alternative strategies to localize DNA circuits. Colloidal systems directed by DNA reaction networks can generate interesting patterns and distributed decision-making behaviours[116–118], which has inspired researchers to integrate DNA logic circuits with colloidal systems to enable particles to perform computational tasks[119,120]. For example, a nanoparticle-based platform was used to construct DNA logic circuits[121] (Fig. 4a). In this example, DNA-functionalized nanoparticles on a supported lipid bilayer serve as computational units. The information storage (memory) and output (reporter) units are immobilized on the lipid membrane, whereas the processing (floater) unit can diffuse freely and bind to immobile particles through DNA hybridization. The state of the memory unit depends on the DNA input presented in the solution, which in turn instructs the behaviours of the processing unit and eventually generates a scattered light signal as the output on the reporter

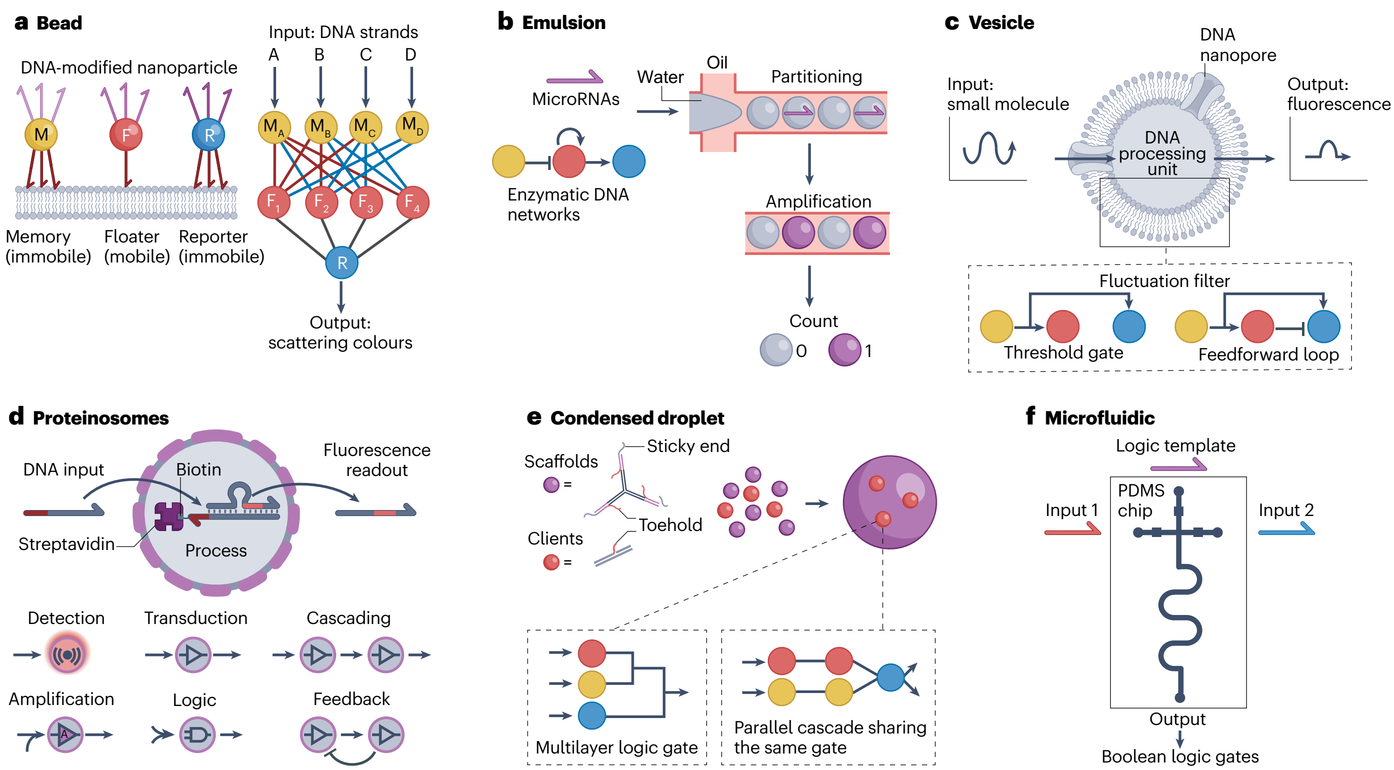

Fig. 4 | Bead-based and compartmentalized DNA computing. a, DNA-modified nanoparticles on a lipid bilayer serve as computational memory (M), floater (F) and reporter (R) units. The memory and reporter nanoparticles are immobile, whereas the floater nanoparticle can diffuse and collide with the other nanoparticles. Addition of DNA inputs changes the hybridization states of the memory unit, and information from the hybridization states is processed by the floater and then reported by the reporter. The binary output is indicated by the binding between the floater and reporter (the output is 1 or 0 for binding or no binding, respectively), which is distinguished by different scattering colours of the nanoparticles. The combination of multiple sets of memory, floater and reporter units enables the programming of a perceptron neural network. **b**, Through partitioning into water-in-oil droplets, an enzymatic DNA circuit can amplify and report the presence of microRNA quantitatively. **c**, Vesicles containing DNA reaction networks can be used to sense, analyse and process the continuous fluctuation of extracellular stimuli. The key module is a fluctuation filter that consists of feedforward loops and threshold gates. **d**, DNA circuits encapsulated within proteinosomes through a streptavidin–biotin interaction can sense, process and secrete a short single-stranded DNA output by DNA strand displacement. Individual proteinosomes can be configured to perform various distributed computing tasks, including detection, transduction, cascading, amplification, logic and feedback. **e**, DNA droplets can function as membraneless compartments to mediate molecular computing. Y-shaped DNA scaffolds (purple circles) with self-complementary sticky ends (that is, unpaired nucleotides that can associate to link DNA segments) condense into dynamic droplets that can host DNA clients (red circles) for various computations. **f**, A microfluidic processing unit chip made from polydimethylsiloxane (PDMS) offers automated operation and programmable control for executing Boolean logic gates. Part **a** adapted with permission from ref. 121, American Association for the Advancement of Science. Part **b** adapted with permission from ref. 130, American Association for the Advancement of Science. Part **c** adapted with permission from ref. 138, American Association for the Advancement of Science. Part **d** adapted from ref. 127, Springer Nature Limited. Part **e** adapted with permission from ref. 148, American Association for the Advancement of Science. Part **f** adapted with permission from ref. 151, American Chemical Society.

module. This sequential strategy allows for the formation of a perceptron neural network. Furthermore, chemical inputs can be processed and transduced into mechanical outputs using motorized particles[122]. In this case, ssDNA on microparticles hybridizes to complementary RNA strands immobilized on a surface. Addition of an RNA nuclease hydrolyses the RNA strands and the particles can then move through a burnt-bridge mechanism. On the basis of this basic element, computing between particles has been established and the final circuit output is the movement of the particles. Instead of fluorophore-encoding, multiplexed particle outputs have been achieved using particles of different sizes and materials. The use of label-free outputs circumvents the limitation of the spectral bandwidth of fluorophores, which could increase parallel computing capabilities. However, the immobilization of DNA circuits on the surface of beads requires careful optimization. As the DNA strands are exposed directly to the environment, the base pairing of the strands needs to be highly orthogonal to each other to minimize unintended crosstalk. The amount of DNA that can be localized on a 2D surface is quite small, which results in slow DSD reactions, especially between particles[122]. In the scenario in which reactive DNA strands are designed to interact with adjacent DNA gates on the same surface, a high grafting density is crucial to prevent loss of local reactants owing to diffusion[116,123].

As an alternative to bead-based strategies, compartmentalization strategies can effectively separate DNA circuits from each other and enable an increase in circuit complexity. With this strategy, the amount of localized DNA is orders of magnitude higher than that achieved through grafting on a particle surface, as an entire compartment is used to localize DNA as opposed to only a surface. Microcompartments such as droplets in emulsions and microfluidics chambers are closed compartments and thus prevent the exchange of materials with the environment. Semipermeable compartments, including vesicles and polymersomes, are permeable to small molecules but can retain large molecules (such as proteins and DNA). However, these compartments can be modified to be permeable to short DNA strands by inserting a nanopore in the membrane of vesicles[124] or by internalizing coacervates inside polymersomes[125]. By contrast, proteinosomes based on self-assembled crosslinked protein–polymer nanoconjugates are inherently permeable to short DNA strands[126,127].

Various strategies for the compartmentalization of DNA circuits have been developed for use in DNA computing. For example, water-in-oil microemulsions with large numbers of DNA circuit-containing droplets with a controlled dispersity are particularly interesting for the screening of parameters that influence the performance of DNA reaction networks. Analysis of more than 10^5 droplets containing an enzymatic DNA reaction network showing oscillatory behaviour revealed that variations in the amplitude, frequency and damping of the oscillations are determined by stochastic partitioning of key components such as enzymes in the droplets, rather than stochastic reaction dynamics[128]. The same methodology has been adapted to produce thousands of droplets with different chemical compositions, yielding high-resolution diagrams that pinpoint the optimal parameters of a reaction network, which is not possible in bulk experiments owing to the extremely large parametric space[129]. Moreover, the small droplet volumes in microemulsions enable DNA computing at the single-molecule level, which can be used for digital sensing with high sensitivity[130,131] (Fig. 4b). The PEN toolbox-based perceptron neural networks (Fig. 2d) have been operated in microemulsions, enabling massively parallel DNA computing[85]. On larger length scales, individual droplets can be connected and organized into multi-compartmentalized networks, enabling genetic circuits with complex spatiotemporal dynamics[132,133].

In another route to compartmentalization, lipid vesicles have been used to encapsulate DNA circuits and to direct the flow of information in response to the external environment[134–137]. The low permeability of vesicular compartments enables efficient retention of DNA reaction networks, whereas the insertion of nanopores in vesicle membranes is a common strategy to facilitate communication by small-molecule signalling. Filtering of noisy information in vesicles has been achieved through the combination of feedforward loops and threshold gates made from DNA[138] (Fig. 4c). These biocompatible sensors can receive molecular cues secreted by live cells in their native biological environment, which could enable continuous biosensing in vivo by biologically programmable biosensors.

Although vesicles can effectively contain molecules without notable leakage, this makes intercompartment chemical communication experimentally more demanding and specific strategies are required to enable information to cross the membrane. An alternative method is to use highly permeable compartments, such as proteinosomes, which allow short DNA strands to diffuse in and out passively. A general and extendable platform that allows for intercompartment signalling has been established using proteinosomes with internalized streptavidin, which enables the selective localization of membrane-permeable biotinylated ssDNA or ssDNA hybridized to a biotinylated partner. Using DSD, non-biotinylated strands can be released or trapped inside the proteinosomes, allowing for programmable messaging between compartments[127] (Fig. 4d). The platform can perform various communication operations, including cascaded amplification, bidirectional communication and distributed computational operations between populations of proteinosomes. In addition, proteinosomes have shown to be compatible with enzymatic DNA reactions such as the CRISPR–Cas system[139] and PEN toolbox[140] for communication operations. So far, compared with emulsion and vesicle systems, only relatively simple DNA circuits have been used inside proteinosomes, but we anticipate that proteinosomes could serve as a platform to perform highly parallel computation with more complex DNA circuits.

It is also possible to achieve compartmentalization in the absence of a confining membrane through immobilization of DNA circuits on a supporting matrix. For example, synthetic polymeric hydrogels functionalized with DNA circuits are capable of changing shape or pattern transformation upon addition of external DNA strands as a stimulus[141,142]. Moreover, membraneless DNA droplets can be formed using long DNA polymers[143–145] or short DNA strands[146]. The structure and organization of DNA droplets can be modulated by simple one-step DSD reactions and more complex logic operations[147]. DNA droplets can also serve as scaffolds to mediate DSD and logic gate operations[148] (Fig. 4e). This method accelerates the computing speed by concentrating reaction components within the membraneless compartment and improves the orthogonality of DNA computation by separating DNA computing from the condensation processes. Additionally, highly charged synthetic polymers or peptides can form membraneless coacervates[149] that serve as scaffolds for DSD[125]. The rich multiphase behaviour of synthetic coacervates might facilitate DNA computing, for example, by providing a multiphase compartment to sort nucleic acids based on their hybridization states[150].

As the scale of DNA circuits increases, DNA computing becomes experimentally more laborious owing to the required manual mixing of strands. To simplify this process, a microfluidic technology has been developed for the automated operation of DNA computing with programmable control[151] (Fig. 4f). The polydimethylsiloxane-based processing unit contains motor-operated valves that can be controlled by a computer or smartphone, enabling the convenient execution of a combination of logic operations. Because molecular circuits make decisions in response to both the presence and history of molecular environments, this prototype could be further developed by integrating DNA-localized chips[152,153] to maintain positional information encoded in diffusive molecular gradients.

Compartmentalization has yet to be widely used as a strategy for DNA computing, but the systems discussed earlier showcase the potential utility of confined media to unlock a higher level of complexity in DNA computing. In the case of non-interacting compartments, the large number of individual computing units could allow for the implementation of stochastic algorithms whereby each compartment returns a unique outcome that is dependent on the slight variation in initial conditions. Conversely, semi-permeable compartments that localize DSD-based computing gates can enable more design flexibility by preventing unwanted interactions between gates[127], and this feature could be further exploited to facilitate the construction of large computing circuits in which it is crucial to prevent unintended crosstalk. Furthermore, although physical training in well-mixed DNA circuits is difficult owing to the lack of spatial information[57], compartmentalization of DNA circuits that act as individual, trainable nodes could allow for the construction of physical learning-based networks based on the local diffusion of signalling strands. In such a system, rather than being a hindrance, diffusion facilitates the local effects needed to train such a network and many identical, simple circuits can be used to construct a computationally more advanced system[154].

Advances in DNA data storage

Parallel to the development of DNA-based molecular computing, the natural function of DNA as an information-carrying molecule has motivated researchers to investigate its use as a synthetic digital data-storage medium. Advances in related biomedical fields, such as genomics and gene synthesis, have spurred developments in this area[155].

There are several advantages of using DNA as a data carrier. First, DNA has been demonstrated to be a robust information carrier with halflives that far exceed those of other digital data carriers (at least thousands of years[29,156] compared with decades[157]), which can be further extended by storing DNA under appropriate conditions[158–162]. Second, because information is stored at the molecular level, the density of DNA-based data storage exceeds that of typical data carriers by approximately six to eight orders of magnitude[163]. Third, the importance of DNA in disease and diagnostics[164] ensures that technologies related to reading and synthesis remain relevant. Finally, nature has provided researchers with a large toolbox of enzymes and other biomolecules that can readily interface with DNA, enabling, for example, facile copying of data[165,166]. In the following sections, we describe methods used for encoding digital data in DNA, the reading of DNA-encoded data and physical organization for increased data densities and longevity, as well as discussing additional methods for increasing data density and, finally, recent success in the post-synthesis editing of DNA-encoded data.

Encoding and reading data in DNA

Encoding of digital data in DNA molecules has been achieved in various ways, which mainly fall into one of two approaches: nucleotide-level encoding (Fig. 5a) or structural encoding (Fig. 5b). We first describe the encoding of digital information (bits) in the sequence of nucleotides that make up a DNA molecule. Theoretically, each nucleotide could store 2 bits of information (for example, using A = 00, C = 01, T = 10 and G = 11 as the encoding). However, this capacity is rarely achieved owing to constraints that prevent certain motifs from being used, the need to balance GC content and the need to prevent long stretches of homopolymers, as these can lead to errors during DNA synthesis and sequencing[167]. Additionally, synthesis limitations require that multiple DNA molecules that are 100–200 nt long are used rather than a single, longer DNA molecule. To ensure appropriate nucleotide distributions in the data-encoding oligonucleotides, various methods have been developed, such as rotating nucleotide tables[30] and codon wheels[29]. Common errors such as insertions, deletions, substitutions or loss of sequences can negatively affect

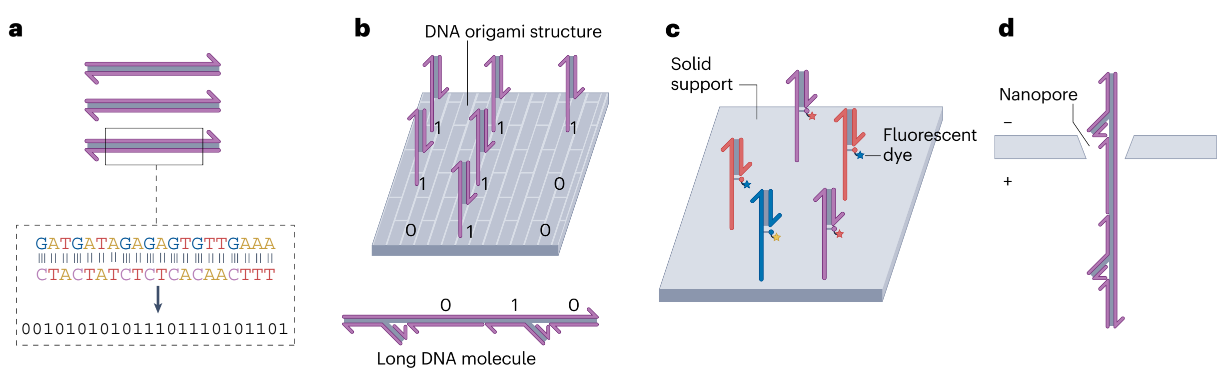

Fig. 5 | Writing and reading information in DNA. **a**, Digital data can be stored in numerous short DNA molecules in which the information is encoded in the nucleotide sequence. The quaternary alphabet of DNA allows for the encoding of more than 1 bit per nucleotide. **b**, Digital data can also be encoded in the structure of DNA molecules. Positional information encoded on DNA origami structures (top) or on long DNA molecules (bottom) can indicate a high (1) or low (0) signal based on the presence or absence, respectively, of a protruding DNA strand. **c**, Digital data encoded at the nucleotide level can be read using sequencing-by-synthesis. Such next-generation sequencing technologies can read many DNA strands in parallel through the localization of sequences on a solid support. These sequences are then used for the step-by-step synthesis of complementary strands (shorter fragments paired to strands to be read). During each step, fluorescently labelled and chemically blocked deoxynucleoside triphosphates (coloured blocks with stars) are added, resulting in single nucleotide extensions that can be fluorescently detected. **d**, Nanopore-based methods can be used to read structurally encoded data. A voltage is applied between the two sides of a solid support in which nanopores are embedded to electrostatically pull DNA structures through the nanopore. Occlusion of the nanopores by DNA nanostructures blocks the flow of ions across the channel; this decrease in flow generates an electronic signature that depends on the size and shape of the DNA structures in the nanopore.

data storage in DNA. Although DNA-encoded data can be protected against these and other errors by physical redundancy – that is, storing many copies of the same sequence – this approach severely hampers the information density of DNA as a storage medium. Therefore, some form of error correction is usually used. The simplest form of error correction for DNA data storage consists of storing multiple copies of the same sequence in overlapping fragments, such that loss of any one sequence does not lead to the erasure of information and redundant segments can be used to correct substitution errors[30]. More advanced error correcting has been realized by using algorithms developed for telecommunication, such as XOR encoding[168], Reed Solomon codes[165,169,170] and fountain codes[171,172]. The direct comparison of different encoding algorithms is challenging, and therefore exact performances are difficult to quantify, but logical data densities of 1.14 bits per nucleotide have been demonstrated[172], and physical densities of 17 EB per gram of DNA have been realized[163] using DNA data storage based on nucleotide-level data encoding. Increasing data densities beyond these benchmarks could be possible using expanded genetic alphabets[173,174] or using degenerate bases to encode information in the distribution of nucleotides at a specific position[175]. Alternatively, data can be encoded in the structure of DNA molecules. Although the data densities that can be achieved using this method are lower than those achieved with nucleotide-level encoding, an advantage is that data can be read more easily. Generally, structure-based encoding has been realized by introducing structural elements such as nicks[176] or bulging structures[177] at predefined positions in a long DNA strand. An advantage of structure-based encoding is that biological DNA sequences can be used as registers[176], which are cheaper to produce than fully synthetic DNA. Additionally, information can be encoded by mapping positions on a flat origami structure to data indices; bits are then stored in the absence (0) or presence (1) of a labelled oligonucleotide at a given position[27].

Reading the data stored in DNA can be achieved by various strategies. Nucleotide-level encoding schemes require sequencing of the DNA molecules. The earliest examples of nucleotide-level encoding used Sanger sequencing to decode the stored data[8]; however, the low-throughput nature of this method prevented practical applications of DNA data storage. Advances in high-throughput and massively parallel sequencing, such as sequencing-by-synthesis[178] and nanopore-based[179] methods, allow for much larger sequence pools to be read and decoded. Owing to the relatively short (100–200 nt) length of typical sequence-level encoded data and lower error rates, sequencing-by-synthesis is the preferred method for reading this type of DNA-encoded data (Fig. 5c). But although nanopore-based DNA sequencing has higher error rates and is more suitable to long-read sequencing, it has been used to sequence DNA-encoded data to demonstrate its versatility[165,180]. The versatility of nanopores is also demonstrated by the ability of this technique to read structurally encoded information[28,181,182], as changes in the excluded volume alter the voltage measured across the nanopores (Fig. 5d). Structural information can similarly be read using methods such as atomic force microscopy or electron microscopy. Atomic force microscopy especially is commonly used to read positional information encoded on DNA origami structures[27,92]. Additionally, structural information can be readout using super-resolution microscopy[27], which can detect unique fluorescently labelled positions on DNA origami structures.

Retrieving and manipulating DNA-encoded data

Despite recent advances, reading of DNA-encoded data is still a costly, time-consuming process and typically not all data need to be read at the same time. Physical separation of the data-encoding DNA into smaller subsets is one method by which excessive reading can be avoided, and various methods for DNA organization have been developed to that end. One example is the organization of distinct dried DNA pools on glass substrates (Fig. 6a); specific DNA-containing spots can be rehydrated using microfluidics and subsequently retrieved[183]. To achieve a higher physical data density, it is preferable to compartmentalize DNA-encoded data into distinct compartments that can be mixed in a single container (Fig. 6b). An early method of compartmentalization for DNA data storage was to store DNA-encoded data inside bacteria to make use of the natural DNA protection and restoration pathways developed by nature[8,184–186]. In this approach, data are encoded in DNA segments that are subsequently integrated into plasmids, which can then be transformed into bacteria. Retrieval of data from bacteria can then be achieved by culturing the bacteria, plasmid purification and, finally, sequencing. As bacterial storage of DNA-encoded data has a relatively low data density compared with storage in neat DNA, synthetic alternatives might offer better performance. Synthetic alternatives could consist of glass particles that encase DNA to separate the data and to increase long-term stability[29,169,170] by shielding the DNA from water and air and thereby stabilizing the DNA for long-term storage[187]. Magnetic beads[166] and microcompartments based on protein–polymer conjugates[188] have also been used to organize DNA-encoded data with the main goal of enabling more reliable and repeated access of the data.

Although the physical organization of DNA libraries is a good strategy to organize very large data sets, more fine-grained control over file retrieval is desirable. Selecting multiple DNA-encoded files by selecting for shared properties of the underlying data, the so-called metadata, has become an increasingly active area of research. For example, tagging compartments with QR codes[189] allows for the inclusion of metadata that describes the compartmentalized DNA pools, but retrieval of compartments from a mixture is hard to automate. Easier automation of data retrieval can be achieved by tagging synthetic compartments with short ssDNA metadata-encoding labels. These labels can subsequently be hybridized with fluorescent labels to enable the selection of specific compartments from larger data pools using fluorescence-assisted cell sorting[188,190] (Fig. 6c). An alternative retrieval method for hybridization-based targeting of DNA-encoded metadata is the use of magnets, and this approach should be easier to automate than fluorescence-based methods. Magnetic retrieval has been used in DNA-encoded data retrieval by labelling magnetic beads with DNA strands complementary to short single-stranded overhangs on DNA files. Similar to the fluorescence-based method, hybridization between the strands allows for selective labelling of the data with magnetic particles. Labelled files can then be magnetically retrieved from the larger library (Fig. 6d). This approach has been used for the amplification of data-encoding DNA by in vitro transcription[191] and similarity-based image search[192]. For single-file-level data retrieval of nucleotide-level-encoded data, the specificity of the PCR is usually preferred over other methods and has been leveraged to enable random access in relatively large DNA data-storage pools[165] (Fig. 6e), with the theoretical potential to scale to terabyte-sized data pools[163]. Building on the success of PCR-based data retrieval, schemes such as nested[193,194], promiscuous[195] and combinatorial[196] PCR have been shown to increase the number of unique files that can be retrieved from a single mixed pool. However, PCR-based random access does not readily scale beyond the, relatively limited, terabyte scale owing to the increase in molecular crosstalk[197,198] between sequences, which negatively affects data retrieval. Although such molecular crosstalk can be prevented through

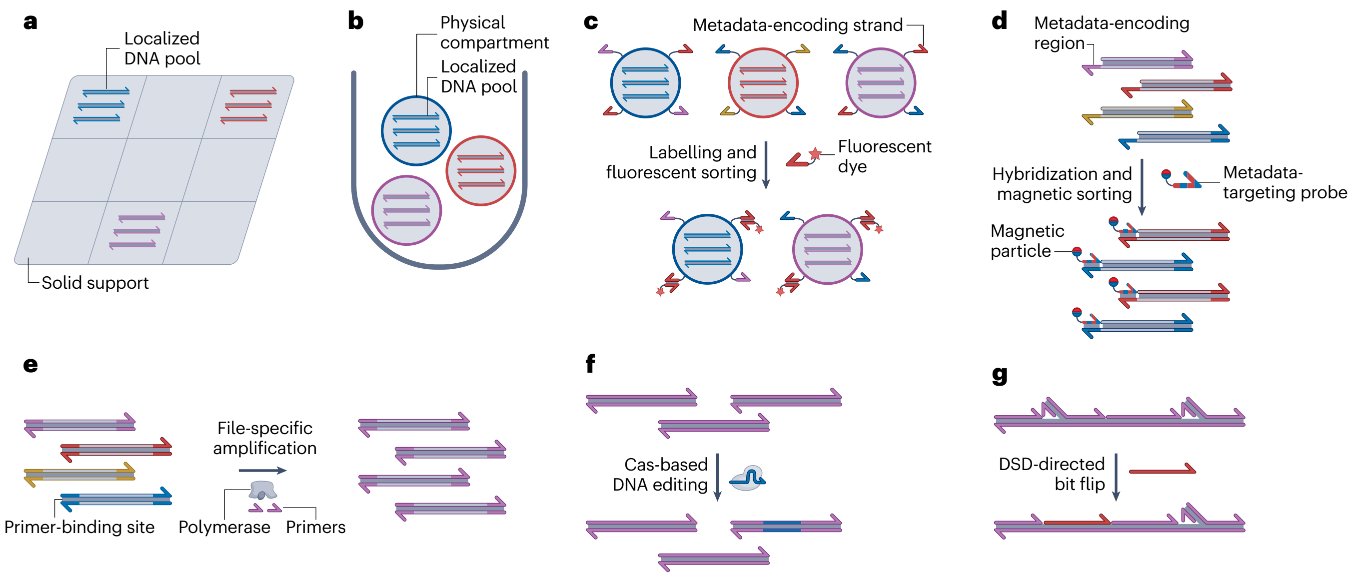

Fig. 6 | Retrieval and manipulation of DNA-encoded data. a, Physical organization of DNA-encoded files on a glass substrate. Different DNA-encoded files that could interact are stored in different areas to ensure reliable data retrieval. This physical isolation is achieved by depositing liquid droplets containing DNA on to the substrate and then dehydrating them. To retrieve the stored DNA, microfluidics is used to move aqueous droplets to the correct spot to rehydrate the stored DNA, which can then be retrieved and decoded. **b**, Compartmentalization of DNA-encoded data. Physical separation in a single container can be achieved by compartmentalizing individual data sets before mixing the compartments together to increase the physical data density. **c**, Fluorescence-based metadata tagging and data retrieval. Compartments containing data-encoding DNA are labelled with short oligonucleotides that represent certain metadata features. To sort the data, a fluorescently labelled DNA strand is added that is complementary to specific metadata strands, thus fluorescently labelling compartments. These compartments can then be fluorescently sorted by fluorescence-assisted cell sorting. **d**, Magnetic file selection based on metadata. Short single-stranded domains of data-encoding DNA strands that describe the data are used for a molecular-level file search. Complementary sequences labelled with a magnetic handle hybridize to the single-stranded domain and are then used to magnetically retrieve the targeted file. **e**, Targeted retrieval of DNA-encoded files by PCR. Single files can be retrieved from a pool of DNA-encoded files using enzymatic amplification, such as PCR, by careful primer design. **f**, Post-synthesis editing of nucleotide-level encoded data. Segments of data-encoding nucleotides can be altered using programmable enzymatic methods, such as the CRISPR–Cas system. **g**, Post-synthesis editing of data encoded in the structure of DNA. Structurally encoded data can be erased or altered after production using toehold-mediated DNA strand displacement (DSD), whereby specific bit-encoding strands can be exchanged to induce a position-specific bit flip (that is, a change from 0 to 1, or vice versa).

compartmentalization of PCR reactions inside emulsion droplets[165,199] or synthetic compartments[188], higher-level preorganization of data is still preferred in practical applications of DNA data storage.

Besides large-scale storage and targeted retrieval, another key component of a file system is that it should allow for the manipulation of stored data – that is, editing of stored data. In principle, editing can be realized by resynthesizing the data-encoding segments to be updated; however, as DNA synthesis is expensive, this approach is undesirable. Editing information encoded in the nucleotide sequences of DNA molecules is challenging; however, it is possible using mutational PCR protocols[200]. Advances in gene editing based on CRISPR–Cas systems have been used in rewriting DNA-encoded data[201,202] (Fig. 6f). Owing to the challenges of enzymatic fidelity, rewriting structurally encoded data is much simpler than nucleotide-level data editing. Structurally encoded data have been updated by exchanging structural elements using toehold-mediated DSD, which enables the specific and facile exchange of DNA strands[203] (Fig. 6g). Alternatively, enzymes such as ligases and programmable nickases can be used to update structural information[176,204].

Although rapid progress has been made to develop each step in DNA data storage, it is still challenging to integrate all procedures in an automated system owing to the different techniques required at each step[180]. Developments in microfluidics-based storage systems[162,205] have demonstrated excellent potential for the development of an integrated DNA data-storage system with low cost and energy consumption, which could aid the development of an integrated DNA hard drive in the future[206].

Outlook

Owing to its high programmability and biocompatibility, DNA has demonstrated promise as a computing substrate. The broad scope of DNA nanotechnology can endow programmable DNA-based nanostructures and microstructures with abilities that range from molecular computing to data storage. However, some open challenges remain to be addressed before DNA computing can realize its full potential.

The outputs of DNA circuits are primarily measured using optical signals, which inevitably complicate the design of the circuits and limit the number of inputs that can be detected in parallel. Label-free methods[122,207,208] that can detect a larger number of outputs should be further developed to tackle this problem. For example, a DNA-based molecular classifier has been coupled with an electrochemical array to enable the multidimensional analysis of biomarkers, including nucleic acids, proteins and metabolic small molecules[209]. To date, most examples of DNA computing circuits use non-reusable circuit components,

Glossary

Anti-factor
A protein that binds sigma factor and thereby inhibits transcription.

Base editors
A gene-editing approach that uses CRISPR–Cas systems to directly introduce point mutations into DNA or RNA.

Boolean logic
Type of algebra that performs logical operations using truth values (TRUE or FALSE) as signals.

Convolutional neural network
A neural network featuring a sparse topology, and thus low connectivity and complexity, to reduce the number of network connections and weight parameters.

DNA circuits
DNA reaction networks that use the concentrations of DNA as signals to perform computational functions, mimicking electronic circuits.

DNA classifiers
Molecular classifiers that use DNA as an input to distinguish different states of a DNA-encoded system.

Feedforward loops
A motif of DNA circuits in which the input has two pathways (direct and indirect) to control the final output.

Fountain codes
Transformation of a data file into an effectively unlimited number of encoded blocks; the original file can be reassembled by any subset of blocks.

Hopfield neural network
Single-layer neural network in which all neurons are both input and output, and they are connected with equal weights.

In vitro transcription and translation (TXTL)
Reactions that harness the cellular transcription and translation machinery to enable protein synthesis in vitro from natural or synthetic DNA templates.

Near-memory computing
An approach in which computing units are located close to the data memory, thereby reducing data movement.

Neural networks
Interconnected artificial neurons in a layered structure that resembles the biological neural network in the human brain.

Perceptron
A basic building block in neural networks that computes a weighted sum of the inputs and their weights and returns a single output value corresponding to the classification of the inputs provided.

Prime editors
A gene-editing approach that uses CRISPR–Cas systems to introduce changes encoded on DNA templates.

Reed Solomon codes
A block coding algorithm in which a message is split into multiple message words, and the encoding algorithm is performed on each message word without any dependency on previously received message words.

Seesaw gate
A gate complex with toehold domains on both sides, enabling reversible exchange of single-stranded DNA.

Sigma factor
A protein required for the initiation of transcription in bacteria.

Threshold gates
A gate that activates when the signal exceeds a set level.

Winner-take-all neural network
Neural network in which neurons compete for activation; the output of a neuron is ON only if the weighted sum of all binary inputs is the largest among all neurons.

XOR encoding
Encoding of data based on XOR logic operations.

which become permanently unresponsive upon reaching the final equilibrium state. There has been less investigation[210–213] towards the implementation of renewable circuits that could facilitate further development of computation, such as sequential logic and memory devices for use in nanorobotics, biosensing and diagnostics.

As the relative performances of electronic computing and DNA computing have been compared and discussed extensively elsewhere[214,215], we here discuss several directions for developing DNA computing and data storage. Although DNA computing might be more energy sufficient, solving computational tasks requires the transformation of digital information into DNA sequences, which is less efficient than electronic computing in terms of time and cost. However, the biological nature of DNA becomes an advantage compared with electronic circuits in bioengineering applications, as DNA circuits can interact directly with biological systems. Currently, most research has been focused on DNA computing on the cell membrane, and leading-edge intracellular applications use only relatively simple circuits[99,216]. However, developing complex DNA-based devices capable of functioning inside live cells and regulating their behaviours remains challenging. In this regard, the first steps towards intracellular DNA computing require DNA circuits with long-term in vivo stability[217] and target specificity[218]. In addition, both DNA-based computation and data storage suffer from slow operating speeds, especially compared with electronic alternatives. Although approaches such as the design of new architectures and use of microfluidic systems might alleviate some of these issues, it is unlikely that these strategies will fully solve them. DNA computing could potentially outperform electronic computing in parallel computing, as many interactions between different complexes can take place in the same reaction volume to enable high parallelism. However, to reduce the need for an increasing number of separate DNA strands[219], the problem must be defined in such a way that allows for sequences to be reused, potentially through compartmentalization. Additionally, because the number of DNA strands required for computing grows exponentially with the size of the problem, improving DNA synthesis is a primary challenge. Chemical synthesis based on phosphoramidite chemistry has limitations in terms of DNA length, scalability and environmental effects, which could be addressed by emerging synthesis methods such as enzymatic DNA synthesis[220].

Given the rapid advances in DNA computing and data storage, problems such as cost[215,220], crosstalk[40], leakage[33] and bias[221] arise as systems continue to scale in density and complexity. As outlined earlier, compartmentalization provides physically separated environments, which enables DNA operation in small reagent volumes with massive parallelism. We anticipate that this strategy will increasingly have an important role in DNA computing and data storage, not only by improving the performance of DNA circuits but also in the engineering of distributed tasks[222] that are complex and not accessible in bulk conditions, similar to how live cells regulate their functions[223].

Thus far, the fields of DNA computing and data storage have developed alongside each other with limited interaction between them. This gap can be partially explained by the different demands placed by the two applications on the sequences and secondary structures of DNA.

Nevertheless, although near-memory computing has been relatively underexplored, it has been demonstrated using toehold-mediated DSD in the SIMD||DNA (single instruction, multiple data) scheme[224]. Using this approach, a system was demonstrated that, given a method to extend the length of the DNA tape, is Turing complete using only a small number of unique domains. However, the use of multiple domains to encode a single bit reduces the data densities that can be realized with this approach. Additionally, the introduction of ssDNA regions to serve as toeholds could decrease the stability of the data; thus, it remains to be seen whether DSD-based data editing is an optimal approach for DNA-based near-memory computing. Alternatively, enzymatic systems might provide a bridge between the opposing demands of computing and data storage. Data-encoding DNA could be edited using programmable Cas-based editors such as base editors[225] and prime editors[226]. The guide RNA required could be transcribed from genelet-like systems that respond to externally added inputs or, potentially, transcribed data segments.

Bridging the gaps between DNA-based neural networks and DNA-based data storage should be a particularly attractive focus for the broader field in the near future. As DNA computing uses a minimal amount of energy, it might be a viable alternative to electronic neural networks, which have been criticized for their high power consumption. Increasingly, near-memory computing chips are being developed that place memory and computing units close together to increase speeds and reduce power consumption; using DNA, it should be possible to perform all computations in the same volume as the data are stored, further decreasing the distance between memory and computing circuits. However, crosstalk is expected to be a limiting factor. To that end, the various compartmentalization strategies we have highlighted might serve to alleviate this bottleneck. In conclusion, the realization of integrated data processing and storage systems using DNA as the computing substrate could unlock the exciting possibilities of extremely parallelized computation on very large data sets with limited energy consumption.

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